

Individual Reflection

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StudyOS Thesis-Finder • individual submission • University of Tübingen

1. Initial expectation

My initial expectation of the course was that it would be a practical, hands-on course where we build super complex and large systems, with a hard-defined goal from the professor. I've expected tasks like building a full-stack AI Agent with a custom orchestration and memory layer. When we talked about finding the right project and the right problem to solve for multiple lectures, I was a bit confused. I tried to understand the reasoning behind this, and suddenly it clicked. Nowadays it's not so much anymore about the how, but more about the what. I've heard a lot of podcasts and fireside chats from startup founders who said, that their biggest task was to find a problem that is actually needed and not a problem which they thought was a problem. My biggest learning in this course was the actual process of identifying a problem that is worth solving by actually interviewing the user group, designing surveys, and analyzing the results.

2. Main learning

My main learning from the course was the process of finding a problem actually worth solving. In the modern age of AI, the technical aspects are most of the time not the bottleneck, because you have smart agents that work 24/7 and can do a lot of work on their own. The biggest problem nowadays is to find a problem that a lot of people actually have. At the beginning of the course, we thought that the major problem of the "missing actual thesis list" is that there exists no specific app that contains all that information. During our user research, we learned that professors do this on purpose, because most of them want to know their students and try to find a thesis topic together with them. After the user survey, we had a real-world experience of "We are building something that can't succeed because the supplier side (professors) don't want it". This really showed me that most of the time, it is better to first verify your hypothesis before you start building the actual solution. Another major learning point was the deep discussions in the meetings about the shape of the product and what we actually ship. In the past, if you wanted to ship something 99% of the time it was an app. Nowadays, that's not true anymore, because we are flooded by distinct apps with little purpose on our phones. The rise of AI agents really enabled us to build a skill to solve a problem, without the need for a full-blown app.

3. Role and contribution in the group

In my opinion, our group worked in a really balanced and efficient way. Since all of us have different backgrounds and different skill sets, we were able to learn a lot from each other while progressing fast. My biggest contribution to the group was my technical knowledge and experience in building and prototyping applications. During the process I've prototyped the initial full-stack approach as well as the architecture and skill layout of the final application. Since our skill needs a lot of manual testing during implementation I've built a simulation and testing framework to automate those steps. Valentin did the majority of the marketing and user research work, like the professor survey, emails for test users, and analyzing the results. Dominique did a lot of testing and developed ideas on how to improve the tool. His major contribution was the pivot into the no-db solution, which was a huge improvement in terms of maintainability. Overall, I think we worked really well together and complemented each other in a great way.

4. Future learning

My future learning goal is to think more structured about the problem and solution space. The hardest part for me in this course was to come up with a problem that is actually worth solving. In university, most of the time you get a hard goal and then just solve it without really thinking about the problem itself. I want to learn how to structure user research, develop an intuition about market fit, as well as how to promote your product and get actual users using it. Additionally, I want to deepen my knowledge about AI/ML and stay up to date with the latest developments in this field. Another major impact was the deep dive into skills, because I have built multiple of those to simplify my day-to-day work.